What One Fragile Subgroup Is Worth: Enrichment, Comparator Uncertainty, and Probability of Success for a Phase II Trial of a Cardiac Extracellular Matrix Hydrogel

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Code and data: github.com/palashraks-afk/ventrigel-cds

Abstract

Background: The VentriGel first-in-man trial (NCT02305602) treated 15 post-infarction patients with an injectable decellularized porcine myocardial matrix. It reported, after the fact, that left ventricular remodeling improved mainly in patients treated more than twelve months after their infarction. Whether that can carry a Phase II design has not been quantified. **Methods:** No simulated patients were used. The trial's Supplemental Appendix gives every endpoint as a mean with its standard error and n, split by stratum, which fixes the standard deviation exactly through $\mathrm{SD} = \mathrm{SEM}\sqrt{n}$. The transcription was checked by recomputing all 38 published p-values that could be checked, and by rebuilding each pooled mean and variance from its strata. The trial never tested the early stratum against the late one; that test was run here and corrected for the number of endpoints examined. The missing control arm was anchored to eight control-arm estimates from six published sources, and the standard error on each anchor was carried through. Results are given as assurance, which is power averaged over the uncertainty in the effect and in the comparator, then multiplied by the probability that the subgroup effect is real. **Results:** All 38 p-values reproduced, with a median standard-deviation reconstruction error of 3.3% (0.3% for end-systolic volume). One endpoint of nine shows a nominally significant interaction: end-systolic volume, −16.9 mL, 95% CI −32.2 to −1.6, $p = 0.034$. It survives neither Bonferroni nor Benjamini–Hochberg correction, although the strata are balanced on all eight baseline measures and the pattern runs opposite to regression to the mean. Natural history after infarction turned out to depend on the era of the trial and to disagree in sign among acute cohorts, while chronic cohorts are stable. Against anchored comparators an unselected trial needs 1,046 patients where an enriched one needs 92. The chronic comparator, however, carries a standard error of 4.4 mL, about the size of the 7.6 mL effect measured against it; carrying that through lowers assurance at 174 patients from 80% to 72%, caps it at 90.9%, and raises the enrollment for an 80% chance of success from 174 to 406. At even odds on the subgroup effect being real, the chance the programme succeeds is 40%. **Conclusions:** Enrichment turns a trial that cannot be powered into one that can, but the conclusion rests on a fragile interaction and on a comparator estimated from 28 patients. A 2×2 design powered on the interaction costs 440 patients, close to the 406 an 80%-assurance enriched trial needs, and it answers the question a sponsor actually has. The next money is better spent measuring the six-month volume change in untreated chronic post-infarction patients than on more enrollment.

Disclosures

Independent secondary analysis of published summary statistics. Not conducted with, funded by, or endorsed by Ventrix, Inc., the Christman Laboratory, or any trial investigator. No patient-level data were accessed. The author declares no competing interests.

1. Introduction

A Phase I trial that establishes safety leaves a sponsor with a design question rather than an answer: how large must the next trial be? For cardiovascular regenerative therapies the question is often decisive. Phase II costs scale steeply with enrollment [18], success rates from Phase II onward are low [17], and single-asset sponsors frequently exhaust their capital between the two phases.

Restricting enrollment to patients more likely to respond, a strategy known as *enrichment*, is established in regulatory guidance for exactly this situation [10, 11]. What is less often made explicit is that the gain depends on the *structure* of the heterogeneity, not merely its presence, and can be negative.

The usual description is that broad criteria "dilute" an effect. Dilution presumes the added patients contribute a smaller benefit, so a larger trial eventually succeeds. If instead they contribute a benefit of the *opposite* sign, the pooled estimate is driven toward zero and can cross it, and no trial of any size succeeds. The two cases are indistinguishable in a summary table and imply opposite decisions.

The VentriGel first-in-man trial supplies an unusually clean instance of the second case [1]. Fifteen patients with prior ST-segment elevation myocardial infarction and moderate left ventricular dysfunction received transendocardial injections of a hydrogel derived from decellularized porcine myocardial extracellular matrix [2, 3]. The trial was single-arm and not powered for efficacy; one stated purpose was to inform which patients a later trial should enroll.

This paper asks what that finding implies quantitatively, and just as importantly, how much of it survives scrutiny. It does not attempt to predict whether an individual patient will respond, a claim fifteen single-arm patients cannot support and which is separately problematic on statistical grounds [16].

1.1. Correction of a prior version

An earlier version of this project, archived at 10.5281/zenodo.21516443, generated 2,000 synthetic patient records from a hand-written eligibility scoring rule, trained a classifier to predict the resulting label, and reported a test ROC-AUC of 1.00.

That result was circular and the present paper supersedes it. The target variable was a deterministic threshold function of the same features supplied to the classifier, so the model recovered a conditional statement its author had written. Reported feature importances reproduced hand-chosen penalty weights, a described hyperparameter search does not appear in the code, and a reported temporal threshold corresponds to nothing in the generator. The classifier is retained in the repository under `deprecated/` with a written account of the error. This correction is stated prominently because the superseded version carries a DOI.

2. Methods

2.1. Recovering the variance structure

Because $\mathrm{SEM} = \mathrm{SD}/\sqrt{n}$,

$$\mathrm{SD} = \mathrm{SEM}\sqrt{n}, \tag{1}$$

which supplies everything a power calculation requires from the published tables. No quantity was digitized from a figure, imputed, or simulated. Nine endpoints were transcribed with their source table recorded alongside each value.

2.2. Validating the transcription

Check 1. Recomputing every published paired t-test p-value as $t = \mathrm{mean}/\mathrm{SEM}$ on $n-1$ degrees of freedom tests the transcription, Equation 1, and the stated test simultaneously.

Check 2. The pooled cohort is the union of the strata, so its moments follow by the laws of total expectation and total variance:

$$\mu = \textstyle\sum_g w_g\, \mu_g, \tag{2}$$

$$\sigma^2 = \underbrace{\textstyle\sum_g w_g\, \sigma_g^2}_{\text{within}} + \underbrace{\textstyle\sum_g w_g\, (\mu_g - \mu)^2}_{\text{between}}. \tag{3}$$

Published totals were never inputs, so agreement is a genuine and demanding test: the between-stratum term is driven by exactly the separation the rest of the analysis depends on.

2.3. Testing whether the subgroup effect exists

The trial compared each stratum against its own baseline and observed that one reached significance while the other did not. That is not a test of effect modification; two subgroups can fall on opposite sides of $p = 0.05$ without differing from each other, and this is among the most common errors in subgroup reporting [12, 13]. The between-stratum difference was therefore tested directly with Welch's unequal-variance t-test on the change scores.

The test has two limitations, both of which belong here and not in a reader's notes. It compares change scores instead of adjusting for baseline by analysis of covariance; with balanced baselines ANCOVA is the standard and more powerful choice, so the reported p is probably conservative, but ANCOVA requires patient-level data. And the multiplicity family is nine endpoints at the six-month visit; the trial also reported one- and three-month visits, and counting those would enlarge the family and weaken the result further. Six months is used because it is the prespecified secondary-endpoint timepoint.

Both Bonferroni and Benjamini–Hochberg [14] corrections were applied. A reduced denominator is also reported, since ejection fraction is algebraically determined by the two volumes and viable mass and scar fraction partition the same myocardium; it is presented alongside the nominal count, not in place of it, because choosing the smaller denominator after seeing the p-values would itself be a form of selection.

Baseline balance was tested on every reported baseline, since the strata were not randomized against each other. Regression to the mean was assessed by asking whether the stratum with the more extreme baseline moved further from the mean, which regression cannot produce.

2.4. Anchoring the missing comparator, with its uncertainty

The Phase I was single-arm, so the early versus late contrast confounds effect modification with differential natural history. Rather than sweeping an arbitrary range, the control-arm change was anchored to published control and placebo arms measuring the same endpoints over approximately six months in comparable populations: TIME [4], PRESERVATION-I [5] and EMPRESS-MI [6] for acute patients; FOCUS-CCTRN [7] and FOCUS-HF [8] for chronic ones; and, for walk distance, a pooled estimate across 38 placebo arms [9]. Indexed volumes were converted at a body surface area of $1.9\,\mathrm{m}^2$, a choice checkable because it implies a VentriGel baseline index of $78.2\,\mathrm{mL/m}^2$ against $65.0\,\mathrm{mL/m}^2$ in the chronic FOCUS-CCTRN population.

Where two anchors cover the same cell the choice is recorded explicitly, with its reason, instead of being left to an automatic rule. This matters: selecting by sample size would have chosen EMPRESS-MI over TIME for the early stratum and thereby reversed the comparator's sign.

Crucially, an anchor is an estimate. FOCUS-CCTRN's zero comes from 28 patients. Its standard error is not published directly, because the trial reports standard deviations of the baseline and follow-up *levels* rather than of the change, and the change SD depends on a test-retest correlation that is never reported:

$$\mathrm{SD}_\Delta = \sqrt{s_1^2 + s_2^2 - 2r\, s_1 s_2}. \tag{4}$$

Taking $r = 0.85$, the conservative end of the range for serial cardiac magnetic resonance, gives a standard error of 4.4 mL, the same order as the 7.6 mL effect being measured against it. That standard error is propagated through every downstream calculation, and the consequence of omitting it is reported separately because it is large.

2.5. Sizing the trial as a function of enrichment

Let π be the late fraction of the eligible pool and e the late fraction of enrolled patients. For treatment-arm change d_g, control-arm change c_g, and a winner's-curse discount λ,

$$\Delta(e) = \lambda \textstyle\sum_g w_g(e)\,(d_g - c_g), \quad w_{\text{late}}(e) = e, \tag{5}$$

with variance given by Equation 3 at $w_g(e)$. The discount multiplies the treatment-versus-control difference rather than the raw change, since the winner's curse inflates the estimated *effect* of having selected the subgroup. Required n scales as σ^2/Δ^2, so enrichment raises $|\Delta|$ and lowers σ simultaneously. Sample size was solved against the noncentral t. All designs use $\alpha = 0.05$ two-sided and 10% dropout. No single discount is defended; every result is reported across the range 0.4 to 1.0.

A design whose effect points away from benefit still returns a finite n, because n depends on Δ^2. Every design is flagged for the sign of its effect, and cost optimization is restricted to designs favouring treatment.

2.6. Probability of success, not power

Assurance [15] integrates power over the uncertainty in the effect,

$$A(n) = \mathbb{E}_\theta[\,\mathrm{power}(n,\theta)\,], \tag{6}$$

with the expectation over the sampling distributions $\mu \sim \bar{x} + (s/\sqrt{n_g})\, t_{n_g-1}$, $\sigma^2 \sim (n_g-1)s^2/\chi^2_{n_g-1}$, and over the comparator $c \sim \mathcal{N}(\hat{c}, \mathrm{SE}_c^2)$. Assurance is evaluated with common random numbers across sample sizes so the curve is exactly monotone.

Assurance is still conditional on the subgroup effect being real. Because that condition is doing substantial work here, the unconditional probability

$$P(\text{success}) = P(\text{effect real}) \times A(n) \tag{7}$$

is reported across a range of priors. No attempt is made to derive the prior; it is a judgement, it is the most consequential single number in the analysis, and the only defensible treatment is to make it explicit and show its consequence.

2.7. Confirming the claim, not only the effect

An enriched trial enrolls only late patients, so it can establish that the therapy works in that stratum but never that timing matters, which is the claim being acted on. A 2×2 design addresses it. With four equal cells the interaction estimate $(\bar{y}_{LT} - \bar{y}_{LC}) - (\bar{y}_{ET} - \bar{y}_{EC})$ has variance $4\sigma^2/n_{\text{cell}}$, twice that of a two-arm comparison at the same per-group n, so required enrollment carries a factor of two relative to a main-effect design of the same effect size. That penalty is partly offset because the contrast exceeds the late-stratum effect alone.

2.8. Screening, cost, and uncertainty propagation

Randomizing N patients at level e from a pool of composition π requires screening $S = \max(eN/\pi,\, (1-e)N/(1-\pi))$. Unit costs are planning assumptions; the primary claims are sample-size ratios, which do not depend on them. Estimation uncertainty is propagated by a parametric bootstrap re-solving the design on each of 4,000 draws, with sign-reversing draws retained as infeasible rather than discarded. Simulation is used only to propagate uncertainty already present in the published estimates or to map assumptions; never to create observations.

3. Results

3.1. The transcription is faithful

All 38 checkable published p-values were reproduced at the precision the source prints. Four cells were excluded for documented reasons: three B-type natriuretic peptide cells, whose table reports percent change while its t-test column tests absolute values, and total-cohort scar fraction, where the printed statistics imply $p = 0.80$ rather than the printed $p = 0.58$. This is an unresolved discrepancy in the source, recorded and not corrected.

Reconstructing each pooled cohort from its strata gave a median standard-deviation error of 3.3%. For end-systolic volume the reconstruction returned −0.36 mL against a published −0.35 mL and 14.26 mL against 14.22 mL, an error of 0.3%.

3.2. One fragile interaction

Figure 1 gives the central inferential result. Exactly one of nine endpoints shows a nominally significant interaction: end-systolic volume, −16.9 mL (95% CI −32.2 to −1.6, $p = 0.034$). Ejection fraction ($p = 0.13$), viable mass ($p = 0.17$) and every remaining endpoint ($p > 0.33$) do not.

It survives no correction. Bonferroni across nine tests requires $p < 0.0056$; Benjamini–Hochberg at $q = 0.05$ rejects nothing. Collapsing the algebraically linked endpoints to five families gives a threshold of 0.010, which $p = 0.034$ still fails.

Two checks favour the finding. The strata are balanced on all eight reported baseline measures (minimum $p = 0.46$; end-systolic volume 156.8 vs 142.3 mL, $p = 0.68$). And the pattern runs opposite to regression to the mean: the early stratum began with the *higher* end-systolic volume and got worse.

This is a suggestive but fragile signal. It is adequate to justify designing a trial around, and not adequate

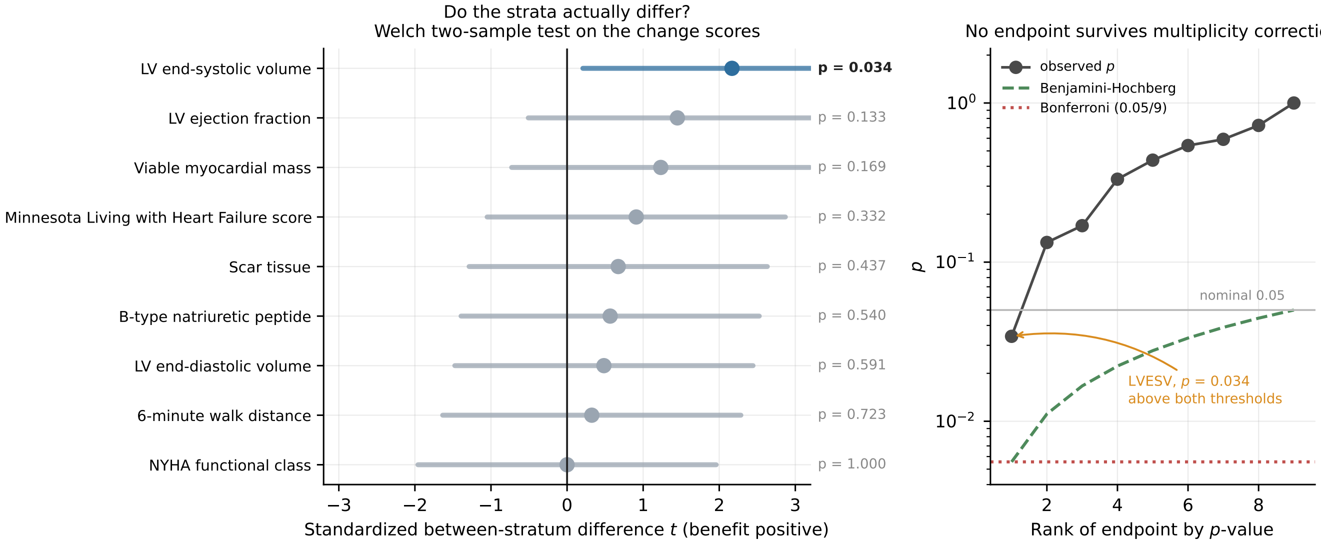

Figure 1: Left: the between-stratum comparison the trial did not perform, standardized so endpoints on different scales share an axis. Right: observed p-values against both multiplicity thresholds.

to assert. Everything below is conditional on it, and Section 3.7 prices that condition.

3.3. What untreated patients do

Figure 2 replaces the missing comparator with measurements, and the result is more interesting than a single number. For *acute* patients the anchors disagree in sign: TIME reported placebo end-systolic index rising 4.3 mL/m^2 and PRESERVATION-I saline diastolic index rising 11.7 mL/m^2, but EMPRESS-MI, enrolling 2022–2024 on contemporary therapy, reported end-systolic index *falling* 7.8 mL/m^2 with ejection fraction rising 8.5 points. Post-infarction natural history is era-dependent. For *chronic* patients the picture is stable: FOCUS-CCTRN observed a placebo change of exactly zero, and FOCUS-HF, with ten control patients, −9.9 mL.

Anchoring two further endpoints changes their interpretation qualitatively. Against a placebo arm whose ejection fraction *falls* 1.3 points, VentriGel's late-stratum −0.6 points is a small benefit rather than the null the trial reported. And against a pooled placebo walk-distance gain of +17.6 m, the late stratum's +50.9 m loses a third of its apparent effect.

3.4. The pooled effect is a cancellation

Figure 3 shows why the pooled numbers mislead: for end-systolic volume the early stratum worsened 9.3 mL while the late stratum improved 7.6 mL, giving a pooled −0.35 mL that is two comparable effects of opposite sign rather than one small effect.

3.5. Sample size under enrichment

Table 1 and Figure 4 give the design consequences. For end-systolic volume an unselected trial requires 1,046 patients against 92 enriched, an elevenfold difference.

Enrichment is not uniformly useful, and anchoring

Table 1: Total randomized patients under anchored comparators, $\alpha = 0.05$, 80% nominal power, 10% dropout, $\pi = 0.53$, 25% discount. Endpoints with no published anchor assume no drift and are correspondingly optimistic.

Endpoint	Δ unsel.	N unsel.	Δ enr.	N enr.
LVESV (mL)	2.64	1,046	5.70	92
Viable mass (g)*	2.32	5,230	11.25	70
MLWHFQ (pts)*	5.79	958	11.32	242
6-min walk (m)	27.51	134	25.01	126
LVEDV (mL)	11.24	126	5.70	366
Ejection fr. (%)	−4.03	*none*	0.53	2,282

* no published anchor; assumes no control drift.

sharpens the point. Walk distance now gains almost nothing from enrichment (1.1-fold), because the placebo response applies to both strata. Ejection fraction remains unusable unselected but becomes technically detectable enriched at 2,282 patients. End-diastolic volume is the one endpoint where enrichment is actively *worse*, because the acute comparator's large positive drift makes the unselected contrast bigger than the enriched one. The governing quantity is the effect in the stratum being excluded, relative to its own comparator.

3.6. The comparator's own uncertainty roughly doubles the trial

Figure 7 isolates a step that is easy to skip. The chronic comparator is 0.0 ± 4.4 mL. Treating it as exact gives 80% assurance at 174 patients and a ceiling of 97.5%. Propagating it gives 72% assurance at the same 174 patients, a ceiling of 90.9%, and **406 patients** for 80% probability of success.

The body-surface-area assumption turns out not to matter for the point estimate, since the late anchor is exactly zero at every value in the plausible range, but

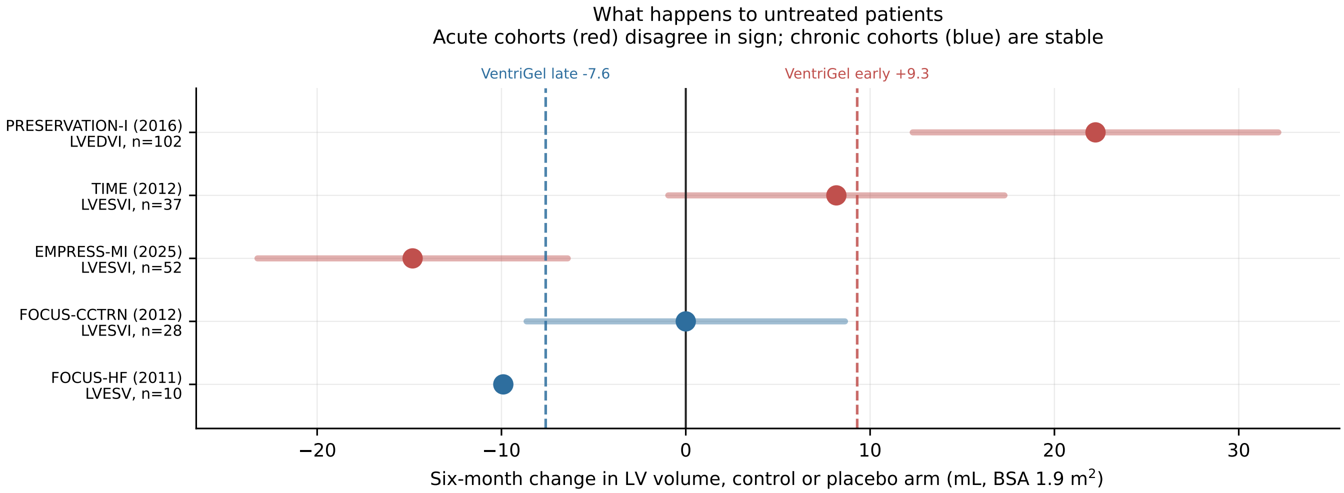

Figure 2: Control and placebo arms of five randomized trials for LV volumes, converted to absolute units. Acute cohorts disagree in sign; chronic cohorts are stable. Dashed lines mark VentriGel's own observed stratum changes.

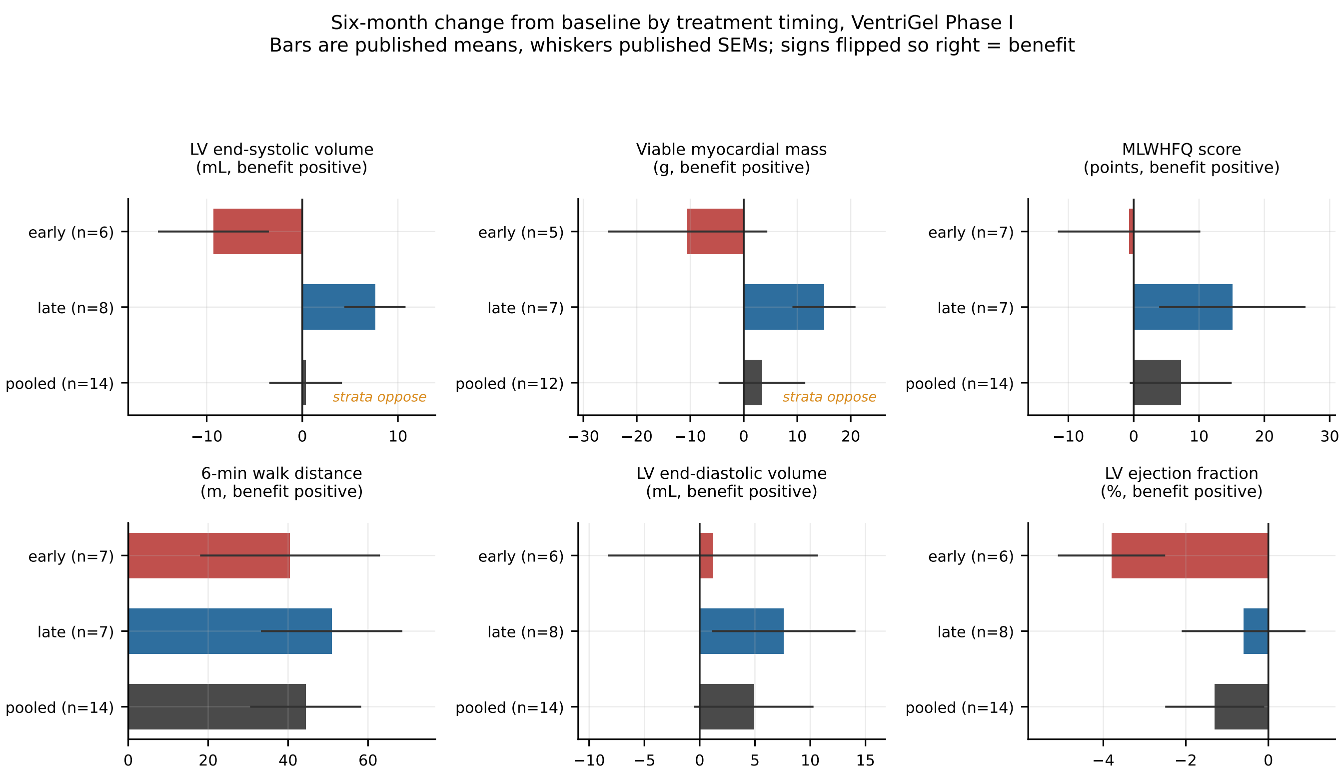

Figure 3: Six-month change from baseline by treatment timing. Bars are published means, whiskers published standard errors, signs flipped so right is benefit.

it moves the standard error from 4.06 to 4.76 mL. The unpublished test-retest correlation matters more, moving it from 2.74 mL at $r = 0.95$ to 6.10 mL at $r = 0.70$.

3.7. Power, probability of success, and the prior

Figures 7 and 8 carry the paper's most consequential correction. The 92-patient design solved for 80% nominal power has an assurance of 63%. Assurance reaches 72% at 174 patients and 80% at 406, and is bounded above by 90.9%.

Multiplying by a prior on the subgroup effect being real changes the picture again. At an even-odds prior the 406-patient design has an unconditional probability of success of 40%, and no sample size lifts the programme above the prior itself. Beyond roughly 400 patients the marginal return on enrollment is close to nothing. That is the quantitative case for spending the next increment of money on the comparator rather than on patients.

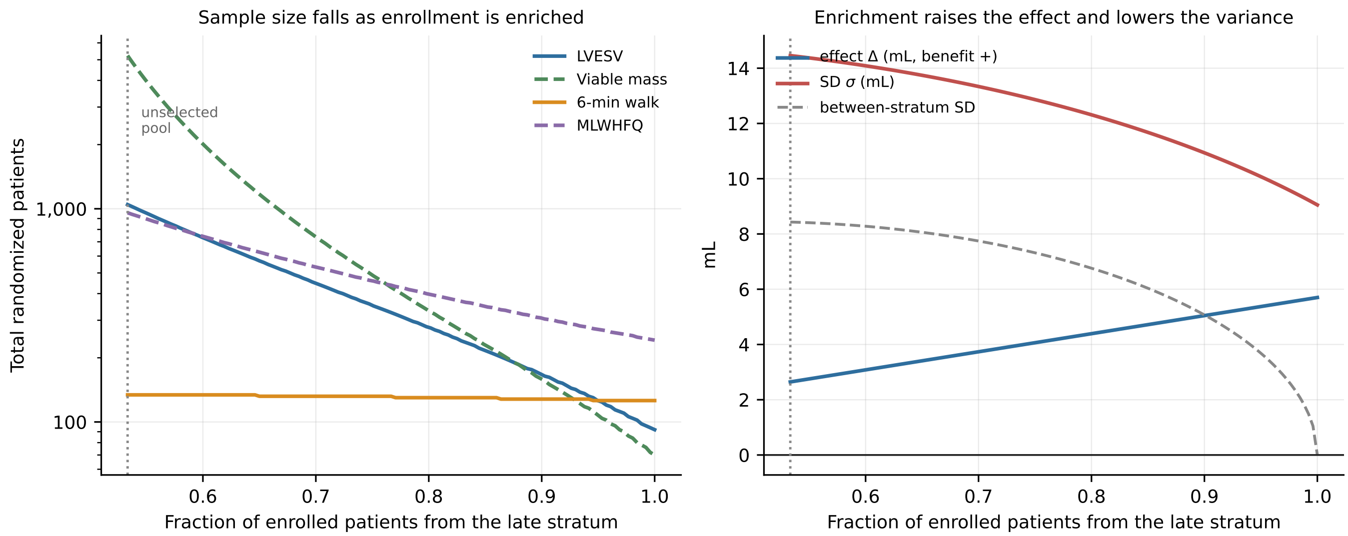

Figure 4: Left: required sample size against enrichment level. Right: the mechanism, which is that enrichment raises the effect and lowers the standard deviation at once.

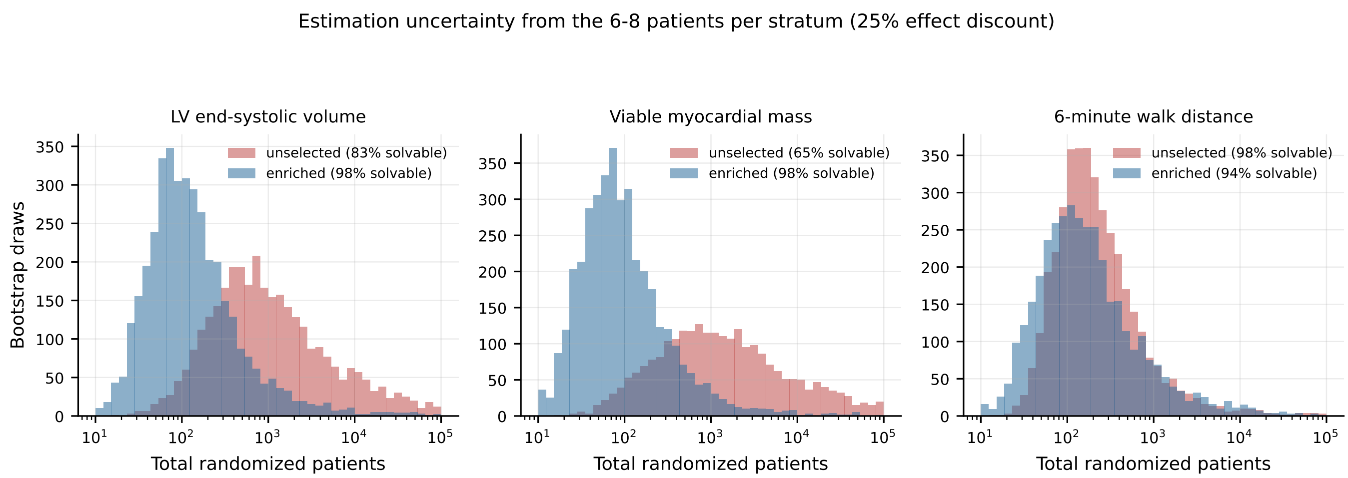

Figure 5: Bootstrap distributions of required enrollment. Intervals are wide because six to eight patients per stratum cannot pin a sample size precisely.

3.8. Confirming the claim costs about the same

An enriched trial cannot establish that timing matters. Powering the interaction instead requires 124 patients against no assumed drift, but only 440 against anchored comparators, because anchoring halves the interaction contrast from 12.7 to 6.5 mL: most of the early stratum's apparent harm is natural history rather than a failure of treatment.

That 440 is close to the 406 an 80%-assurance enriched trial needs. For approximately the same enrollment a sponsor can answer the question they actually have.

3.9. Estimation uncertainty and feasibility

Figure 6 maps the full assumption space. The enriched design enrolls no early patients, so its own size is unaffected by the early comparator; it depends entirely on the late one. Under FOCUS-CCTRN's zero the design stands; under FOCUS-HF's −9.9 mL, VentriGel's −7.6 mL is smaller than natural history and no trial is worth running. Figure 5 reports estimation uncertainty; the enriched end-systolic volume design has a bootstrap median of 108 with an 80% interval of 34 to 635.

Under the stated cost assumptions the 406-patient recommended design requires 761 screens and approximately \$42M, and needs 38 sites to complete in 36 months against the six that ran the Phase I. The screening penalty binds harder as responders thin: enrollment stays fixed but the screens needed to find that enrollment scale as $1/\pi$, so recruitment duration rather than sample size becomes the binding constraint in a scarce population.

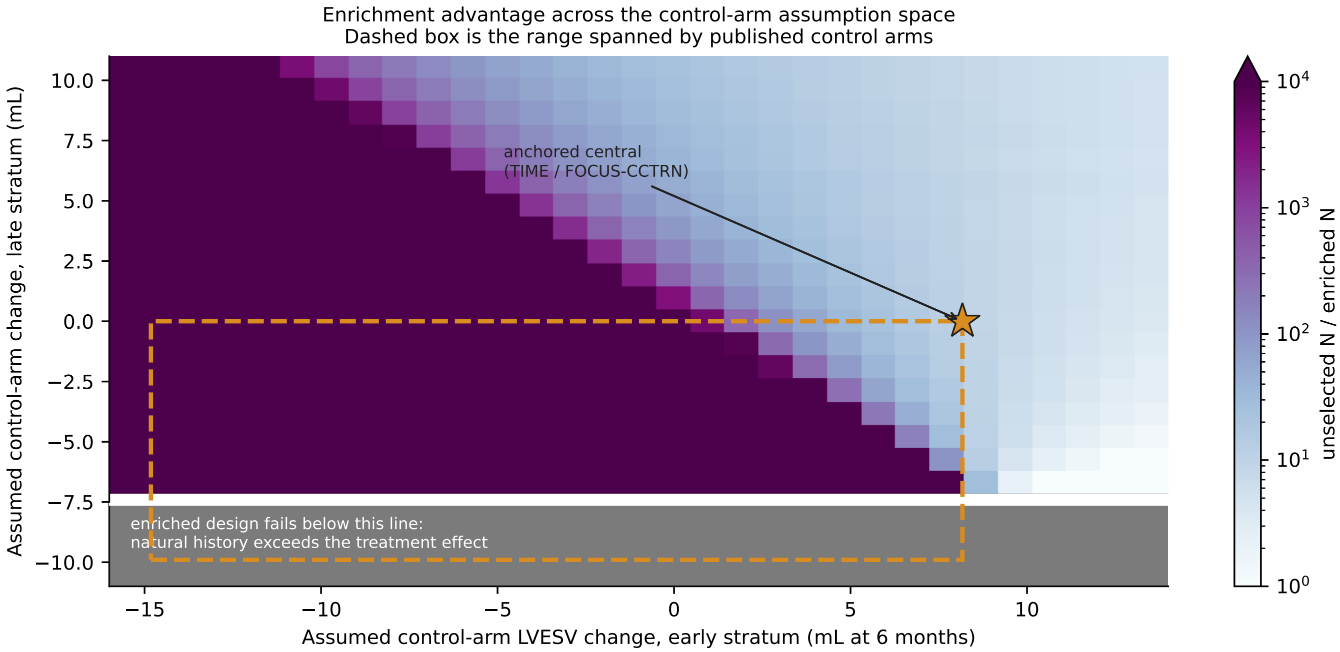

Figure 6: Enrichment advantage across the comparator assumption space, as the ratio of unselected to enriched enrollment. The top of the colour scale means the unselected trial has no benefit to detect at any size. Below the white line the enriched design fails outright, because assumed natural history exceeds the treatment effect. The dashed box spans the range of the published control arms and the star marks the anchored central case.

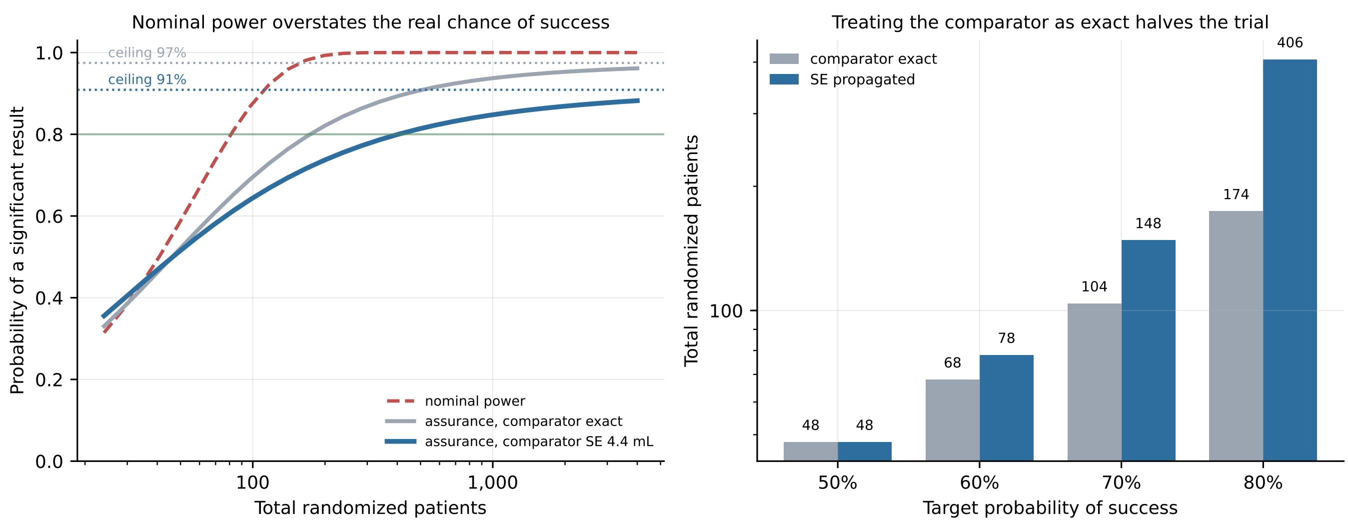

Figure 7: Nominal power against assurance, and the enrollment required for a target probability of success.

4. Discussion

4.1. What a sponsor should do

If the reported pattern is real, a Phase II powered on left ventricular end-systolic volume should restrict enrollment to patients more than twelve months post-infarction and plan for approximately 400 randomized patients across roughly 38 sites. But the same enrollment buys a 2×2 design that also establishes whether timing matters, and that is the better purchase.

Before committing to either, a sponsor should measure the six-month end-systolic volume change in untreated chronic post-infarction patients with LVEF $\leq 45\%$. That single quantity determines whether the effect exists at all, is currently estimated from 28 patients with a standard error comparable to the effect itself, and is obtainable from existing observational cohorts with serial cardiac magnetic resonance. It is a far better use of the next increment of money than additional enrollment, whose marginal return is close to zero beyond 400 patients.

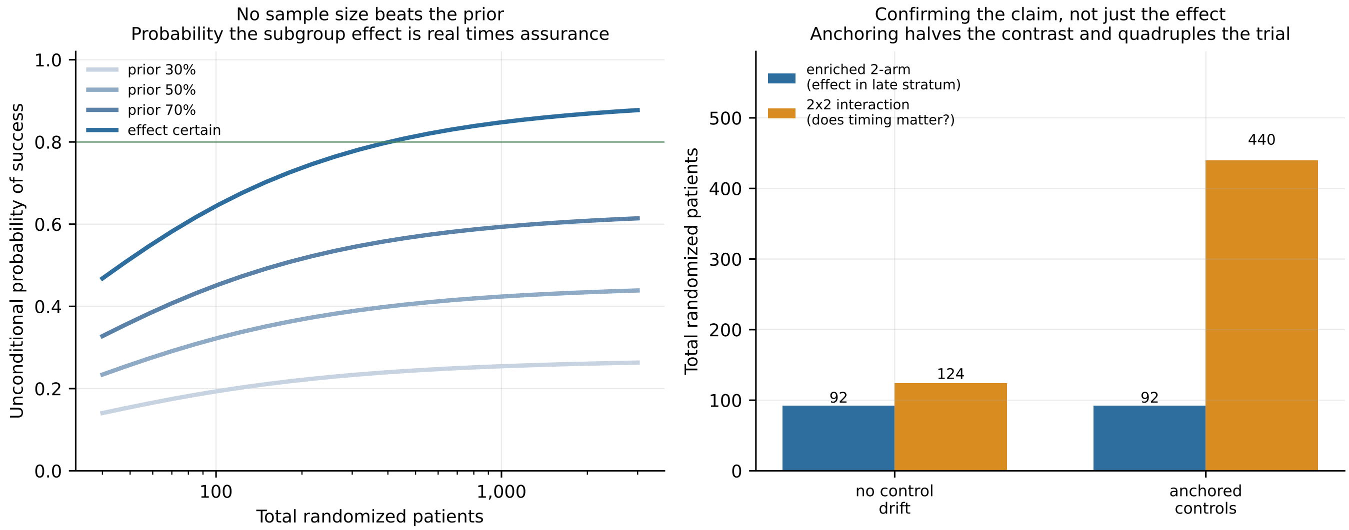

Figure 8: Left: unconditional probability of success is the prior on the subgroup effect times the assurance, so no sample size beats the prior. Right: the cost of confirming that timing matters, rather than only that the therapy works in one stratum.

4.2. The transferable points

Three generalize beyond this therapy. First, describing heterogeneity as "dilution" obscures the decision: dilution implies a recoverable effect and a larger trial, cancellation implies neither, and distinguishing them requires testing the interaction rather than reading two within-subgroup comparisons. Second, an externally borrowed comparator is an estimate, and its standard error belongs in the design calculation; here omitting it halved the apparent trial. Third, when a design rests on a subgroup claim, assurance conditional on that claim is not the number a sponsor needs.

4.3. Limitations

The interaction that motivates the analysis survives no multiplicity correction. It is one nominally significant result among nine, from a post hoc subgroup, in a fifteen-patient single-arm trial, and tested on change scores rather than by ANCOVA. The strata contribute six and eight patients for the imaging endpoints.

External anchoring improves on an arbitrary sweep but substitutes different populations for the missing control arm; the acute anchors are days to two weeks post-infarction while the early stratum is 60 days to twelve months, and they disagree in sign. Body surface area and the test-retest correlation were both assumed. The walk-distance anchor is a conference abstract rather than a peer-reviewed report and is labelled as such. Viable mass and quality of life have no anchor at all and their figures in Table 1 are correspondingly optimistic.

Cardiac magnetic resonance follow-up was incomplete, so stratum n varies by endpoint. The prevalence of late-stratum patients in a real eligible pool is unknown. Unit costs are planning assumptions.

Finally, statistical detectability is not clinical importance. A 7.6 mL reduction on the late stratum's 142.3 mL baseline is a 5.3% change, modest against the reverse-remodeling thresholds usually considered clinically meaningful. Powering for a clinically motivated target would increase every figure in Table 1.

5. Reproducibility

All results are generated by `python run_analysis.py` and `python make_figures.py` with no arguments and no downloaded data. Published values are transcribed in `ventrigel/trial_data.py` and external anchors in `ventrigel/literature.py`, each with its source. The validation of Section 3.1 and the inference of Section 3.2 run on every execution, so a transcription error surfaces as a failed check rather than a silently wrong result. A test suite covers the power mathematics against textbook reference values and pins the headline claims.

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